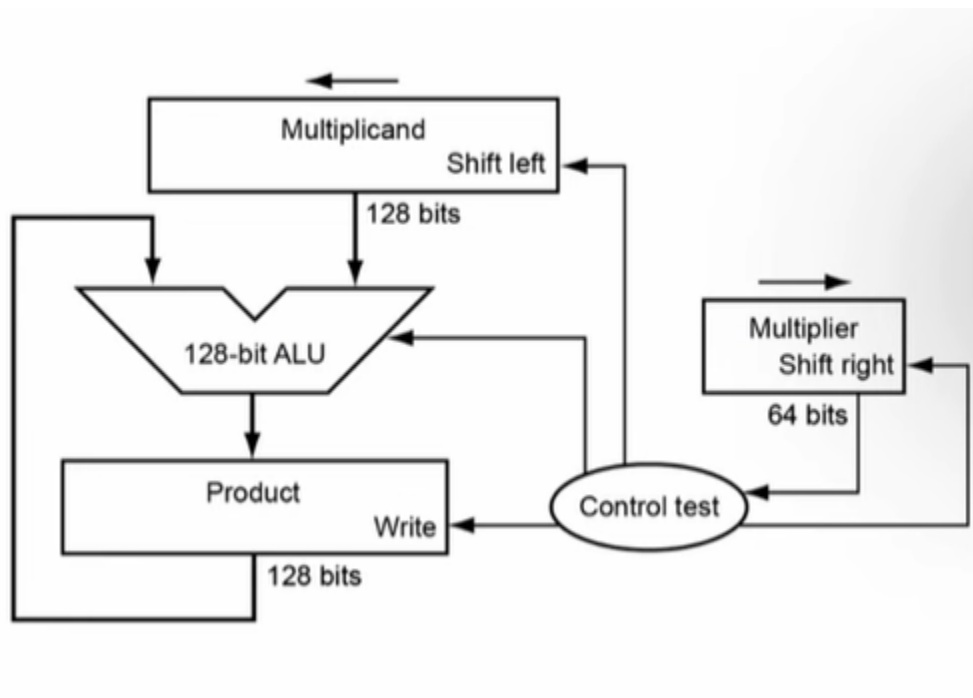
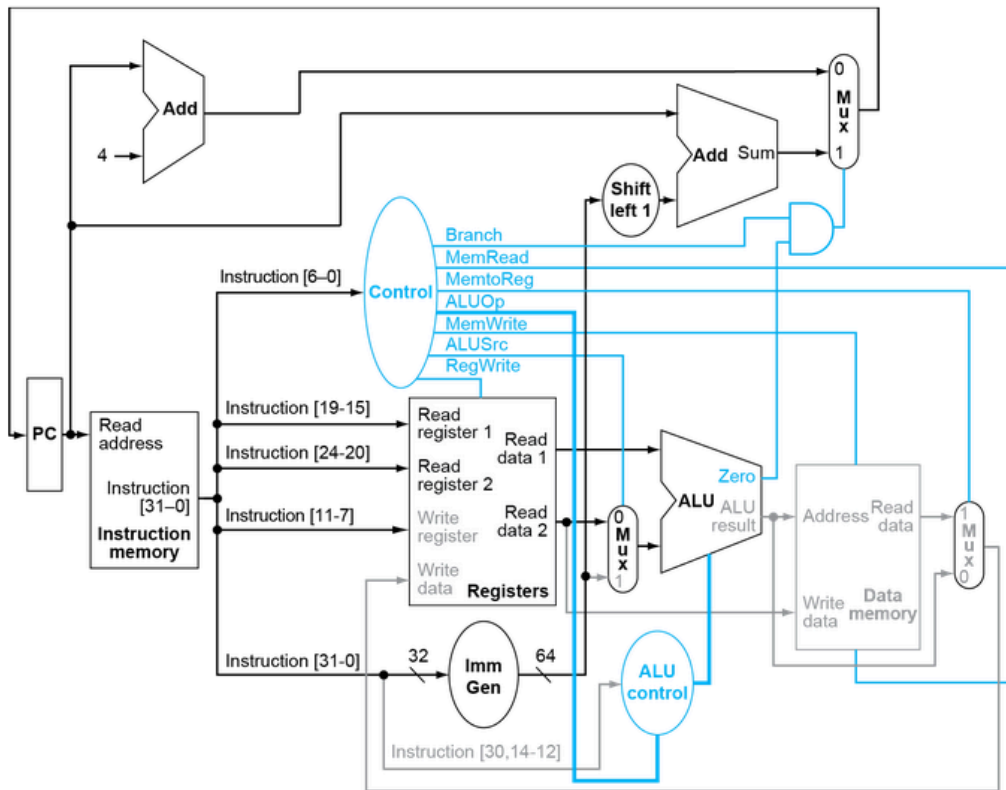


Chapter 2

Chapter 3



Chapter 4



2. Suppose we run the following instruction in a single cycle RISC-V datapath.

Lb x5, 16(x8);

Now answer the following questions based on that: [4 x 1]

- Which component is responsible for calculating the address? Answer: **ALU**
- Name two datapath components that are used by this instruction but not used by any R type instruction? Answer: **Data Memory, Immediate Generation Unit.**
- What will be the value of the Branch control bit? Answer: **0**

3. Explain Why does the Immediate Generation unit take the entire 32-bit instruction as input instead of only the immediate field, and why does it generate a signed output? [5]

Answer:

Different RISC-V instructions use different immediate sizes.

- Load, Store, Beq, Bne, Blt, Bge use 12-bit immediates.
- Jal and Lui use 20-bit immediates.

The position of the immediate bits is not fixed across instruction formats; the bits are located in different places depending on the instruction type.

Hence, it takes the entire 32-bit instruction as input and uses the opcode to determine the size and layout of the immediate field.

addi x5, x7, -8 requires a negative immediate, and branch instructions need signed offsets to allow forward and backward jumps.

1. Why does a Register File require two read reg ports and only one write reg port? [5]

Answer:

Format	rs	rd	Across all RISC-V instruction formats, an instruction can have at most two source registers and at most one destination register . Since no instruction needs more than two registers to be read simultaneously and no instruction writes to more than one register at a time, therefore the register file is designed with two read ports and one write port.
R Type	2	1	
I Type	1	1	
S Type	2	0	
U Type	0	1	
SB Type	2	0	
UJ Type	0	1	

- ALU used for
 - Load/Store: F = add
 - Branch: F = subtract
 - R-type: F depends on opcode

ALU control	Function
0000	AND
0001	OR
0010	add
0110	subtract

2. A CPU can execute only one instruction in a clock cycle.

Instruction	Latency
Add	8s
Sub	10s
Ld	15s
Sh	13s

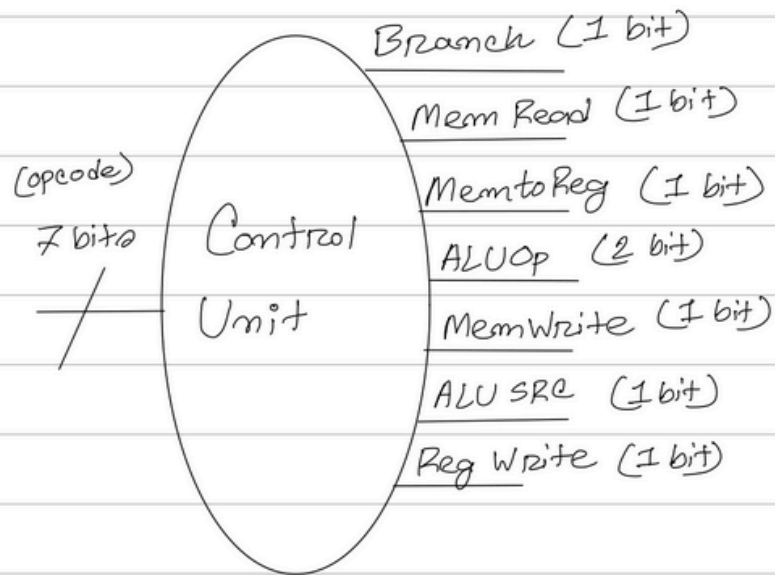
Suppose we run the above mentioned instruction set in the CPU. Now answer the following questions based on that: [4 x 1]

- Determine the clock period of the CPU. Answer: 15s
- How many clock cycles are needed to run the instruction set? Answer: 4
- What is the time taken to run the Sh instruction in that CPU? Answer: 15s
- If we remove the Sh instruction from the instruction set, would the clock period change?
Answer: No

2. Why do we use an ALU Control unit instead of controlling the ALU directly from the main Control Unit? [5]

Answer:

Control Unit takes the opcode as input and uses it to detect the instruction format. For R type instructions, we can't detect what is the exact instruction just using the opcode values. If we can't detect the exact instruction then we can't determine the ALU operation for that instruction. Funct3 and Funct7 fields are used to detect the exact instruction for R type instructions. Therefore, a separate ALU Control Unit is used to combine the ALUOp signal from the main Control Unit with the funct fields to generate the actual ALU control signals which tells the ALU to perform correct operations.



ALU Op $\Rightarrow$ 00 $\rightarrow$ load / Store / Addi

01 $\rightarrow$ Beq

10 $\rightarrow$ Rtype

Bonus Content:

How to detect Hazard:

- (i) Ignore the first line.
- (ii) For the current instruction, list its ^(rs) source registers.
- (iii) Compare these sources only with the ^(rd) destination registers of the previous two instructions.
- (iv) If no match → No Hazard.
- (v) If match → Data Hazard exists.
- (vi) If both previous instructions update the same register, use the latest update.

Hence, A stall is required only when a load is immediately followed by an instruction that needs the loaded value in the Ex Stage.

ALU Control

↳ it controls ALU's operations.

this input works only when ALUOp is 10

imm [30, 14-12]

ALUOp

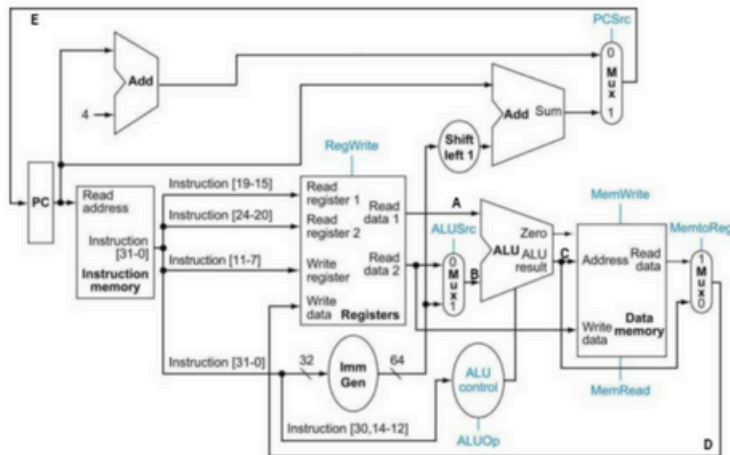


4 bits

- ↳ 0000 → AND
- ↳ 0001 → OR
- ↳ 0010 → ADD
- ↳ 0110 → Sub

opcode	ALUOp	Operation	ALU function	ALU control
ld	00	load register	add	0010
sd	00	store register	add	0010
beq	01	branch on equal	subtract	0110
R-type	10	add	add	0010
		subtract	subtract	0110
		AND	AND	0000
		OR	OR	0001

Question - 17:



Study the above RISC-V Datapath thoroughly. Assume that initially $PC = 600h$, $x_1 = 222h$, $x_2 = 444h$, $x_3 = 666h$, $x_4 = 999h$.

Determine the values of A, B, C, D, E in hex for the instructions executed in following order. If the value does not matter, write it as X (don't care)

		A	B	C	D	E
i.	sub x_1, x_2, x_1	444h	222h	222h	222h	604h
ii.	sd $x_4, 12(x_2)$	444h	Ch	450h	X	608h

Question - 18:

Carefully examine the RISC-V datapath provided in Q-17. Assume the initial values are: $PC = 600h$, $x_1 = 222h$, $x_2 = 444h$, $x_3 = 666h$, and $x_4 = 999h$. For the following instructions executed **sequentially**, **determine** the hexadecimal values of A, B, C, D, and E. If the value is irrelevant, indicate it as X (don't care).

i.	sub x_1, x_2, x_1	try yourself!					
ii.	sd $x_4, 12(x_2)$						
		PC	A	B	C	D	E
iii.	Addi $x_1, x_1, 12$				↑		

The following table shows the different stages involved in executing instructions and the corresponding durations for each stage:

stages	Instruction Fetch	Register Read	ALU Op	Memory Access	Register Write
Duration	50ps	10ps	30ps	20ps	10ps

Given the above durations, determine the total time required to complete each of the following instructions:

	Instructions	Time to complete each instructions
i.	ADD X_{21}, X_{22}, X_{23}	$50 + 10 + 30 + 10 = 100ps$
ii.	AND X_{21}, X_{22}, X_{23}	} try yourself.
iii.	OR X_{21}, X_{22}, X_{23}	
iv	ADDi X_{21}, X_{22}, X_{23}	
v.	LD $X_{21}, 22(X_{21})$	
vi.	SD $X_{21}, 22(X_{21})$	
vii	BEQ X_{21}, X_{22}, End	

Question - 22:

Instruction	Time (PS)
Add X21, X22, X23	10
Sub X21, X23, X24	20
Mul X22, X23, X26	15
LD X22, 0(X21)	25

The above instructions are being run in a **single cycle datapath**.

- Now determine what is the clock period of this system?
- What would be the total time to run this instruction sequence?

In a **single-cycle datapath**, each instruction completes in one clock cycle.

Question - 24 :

Stage	IF	ID	EX	MEM	WB
Time (PS)	10	20	20	20	10

The below instructions are being run in a pipelined datapath. Calculate the time required to execute each instruction.

Instruction	Time (PS)
Add X21, X22, X23	
Sub X21, X23, X24	
Mul X22, X23, X26	
LD X22, 0(X21)	

Determine what is the clock period of this system?

In a **5-stage pipelined datapath**, the **clock period** is set by the **slowest stage** (the maximum stage delay).

Given stage times (ps):

IF = 10, ID = 20, EX = 20, MEM = 20, WB = 10

1) Clock period

$T_{clk} = \max(10, 20, 20, 20, 10) = 20$ ps
 $T_{clk} = \max(10, 20, 20, 20, 10) = 20$ ps

2) Time to execute (complete) each instruction

In a pipeline, an instruction takes **one cycle per stage** to finish.

A typical instruction goes through **IF → ID → EX → MEM → WB = 5 cycles** (even if MEM/WB do “nothing” for some instructions, the pipeline stage still exists).

So completion time per instruction:

$$5 \times T_{\text{clk}} = 5 \times 20 = 100 \text{ ps} \quad 5 \times T_{\text{clk}} = 5 \times 20 = \boxed{100 \text{ ps}} \quad 5 \times T_{\text{clk}} = 5 \times 20 = 100 \text{ ps}$$

Therefore (for the table):

- **ADD x21,x22,x23:** 100 ps
- **SUB x21,x23,x24:** 100 ps
- **MUL x22,x23,x26:** 100 ps
- **LD x22,0(x21):** 100 ps

✓ Clock period = 20 ps

✓ Time per instruction (latency) = 100 ps

*(Extra note: once the pipeline is full, the CPU can finish **one instruction every 20 ps** — that's throughput, not*

Question - 31 :

The following are the time requirements for various stages of a datapath:

1. Instruction Fetch: 25 ps
2. Instruction Decode: 10 ps
3. Register Read: 15 ps
4. Arithmetic or Logical task execution: 20 ps
5. Data Memory Read: 20 ps
6. Data Memory Write: 20 ps
7. Register Write: 25 ps

You are given the following instruction set, executed sequentially:

1. Add X21, X22, X23
2. Sub X21, X23, X24
3. Mul X22, X22, X26
4. LD X22, 0(X20)

Suppose this instruction set is executed sequentially in a single cycle datapath.

Answer the following:

- a. **Calculate** the **actual** time required to execute each instruction.
- b. **Determine** the clock period of this system.
- c. **Calculate** the number of clock cycles needed to execute this instruction set.
- d. **Determine** the total time required to execute this instruction set.

We're told it's a single-cycle datapath $\Rightarrow$ every instruction finishes in exactly 1 clock cycle, and the clock period must be long enough for the slowest instruction path.

Given stage times:

- IF = 25 ps
- ID = 10 ps
- Reg Read = 15 ps
- ALU/Logic = 20 ps
- Data Mem Read = 20 ps
- Data Mem Write = 20 ps
- Reg Write = 25 ps

(a) Actual time required by each instruction (sum of stages it uses)

1) ADD x21,x22,x23 (R-type)

Uses: IF + ID + RegRead + ALU + RegWrite

$$25 + 10 + 15 + 20 + 25 = 95 \text{ ps}$$

2) SUB x21,x23,x24 (R-type)

Same stages as ADD:

$$25 + 10 + 15 + 20 + 25 = 95 \text{ ps}$$

3) MUL x22,x22,x26 (assume it's an ALU-type instruction here)

Also uses: IF + ID + RegRead + ALU + RegWrite

$$25 + 10 + 15 + 20 + 25 = 95 \text{ ps}$$

4) LD x22, 0(x20) (Load)

Uses: IF + ID + RegRead + ALU(address) + MemRead + RegWrite

$$25 + 10 + 15 + 20 + 20 + 25 = 115 \text{ ps}$$

✓ So:

- ADD = 95 ps
- SUB = 95 ps
- MUL = 95 ps
- LD = 115 ps



(b) Clock period of the single-cycle system

Single-cycle clock period = **time of slowest instruction** (max actual time):

$$T_{clk} = \max(95, 95, 95, 115) = \boxed{115 \text{ ps}}$$

(c) Number of clock cycles needed for the instruction set

Single-cycle $\Rightarrow$ **1 cycle per instruction**.

There are 4 instructions:

$$\text{cycles} = 4 \times 1 = \boxed{4 \text{ cycles}}$$

(d) Total time to execute the instruction set

Each instruction takes **one full clock period** (even if it could finish earlier).

$$T_{\text{total}} = (\text{#instructions}) \times T_{\text{clk}} = 4 \times 115 = \boxed{460 \text{ ps}}$$

Final Answers

a) ADD = 95 ps, SUB = 95 ps, MUL = 95 ps, LD = 115 ps

b) Clock period = 115 ps

c) Clock cycles = 4

d) Total time = 460 ps

Instruction Types (examples)	RegWrite	MemRead	MemWrite	ALUSrc	Branch	MemtoReg
R-type: add, sub, and, or etc.	1	0	0	0	0	0
Load: lw	1	1	0	1	0	1
Store: sw	0	0	1	1	0	X
Branch: beq	0	0	0	0	1	X

Instruction	RegWrite	MemRead	MemWrite	Branch	MemtoReg
Bne X3,X4, Else	0	0	0	1	x
Ld X3, 32(X4)	1	1	0	0	1
Sd X3, 32(X4)	0	0	1	0	0
Addi x5, x6, 9	1	0	0	0	0

$$\text{CPU time} = \text{Instruction count} \times \text{CPI} \times \text{Clock Cycle Time}$$

Determined by the

Determined by

language or platform

CPU hardware.

you are using. → ISA, Compiler

→ How many Risc-V instructions we need to complete the task/code?